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student@student-HP-Pro-T... x student@student-HP-Pro-T...
GNU nano 7.2 filename.
#!/bin/bash

if [ "$EUID" -eq 0 ]
then
    echo "You are the root user."
else
    echo "You are NOT the root user."
fi
```

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student@student-HP-Pro-T... x student@student-HP-Pro-T... x student@student-HP-Pro-T...
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ nano filename.sh
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ chmod +x check_root
chmod: cannot access 'check_root.sh': No such file or directory
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ chmod +x check_root
chmod: cannot access 'check_root.sh': No such file or directory
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ chmod +x filename.sh
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ ./filename.sh
You are NOT the root user.
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$
```

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student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~
student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ NANO DISK_ALERT.SH
NANO: command not found
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ nano disk_alert.sh
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     1.6G       2.4M   1.6G   1% /run
/dev/nvme0n1p6            96G       24G    68G   26% /
tmpfs                     7.7G       0    7.7G   0% /dev/shm
tmpfs                     5.0M       8.0K   5.0M   1% /run/lock
efivarfs                  256K      164K    88K   66% /sys/firmware/efi/efivars
/dev/nvme0n1p1            256M      102M   155M   40% /boot/efi
tmpfs                     1.6G      152K   1.6G   1% /run/user/1000
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ fallocation -l 40G testfile
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     1.6G       2.4M   1.6G   1% /run
/dev/nvme0n1p6            96G       64G    28G   71% /
tmpfs                     7.7G       0    7.7G   0% /dev/shm
tmpfs                     5.0M       8.0K   5.0M   1% /run/lock
efivarfs                  256K      164K    88K   66% /sys/firmware/efi/efivars
/dev/nvme0n1p1            256M      102M   155M   40% /boot/efi
tmpfs                     1.6G      152K   1.6G   1% /run/user/1000
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$
```

```
GNU nano 7.2
#!/bin/bash

THRESHOLD=90
TO="chavan.shounak05@gmail.com"
SUBJECT="Disk Usage Alert - 8AM Check"
HOSTNAME=$(hostname)

# Get root partition usage
USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')

# If usage >= 90%, send email
if [ "$USAGE" -ge "$THRESHOLD" ]; then
    echo "Warning!
Disk usage on $HOSTNAME is ${USAGE}%.
Threshold: ${THRESHOLD}%.
Immediate action required." \
    | /usr/bin/mail -s "$SUBJECT" "$TO"
fi
```

```
chmod: cannot access 'check_root.sh': No such file or directory
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ chmod +x filename
h
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ ./filename.sh
You are NOT the root user.
(base) student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$
```

```
student@student-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~
GNU nano 7.2 /bin/bash
#!/bin/bash

# Database name
DB_NAME="testdb"

# Backup directory
BACKUP_DIR="/home/shounak_2450/mysql_backups"

# Create backup directory if not exists
mkdir -p "$BACKUP_DIR"

# Timestamp
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")

# Backup file name
BACKUP_FILE="$BACKUP_DIR/${DB_NAME}_backup_${TIMESTAMP}.sql"

# Take backup using sudo (auth socket)
sudo mysqldump "$DB_NAME" > "$BACKUP_FILE"

# Check if backup successful
if [ $? -eq 0 ]; then
    echo "Database backup successful: $BACKUP_FILE"

    # Compress the backup file
    echo "Compressing backup file..."
    cd "$BACKUP_DIR"
    zip "$DB_NAME_backup_${TIMESTAMP}.zip" "${DB_NAME}_backup_${TIMESTAMP}.sql"
    mv "${DB_NAME}_backup_${TIMESTAMP}.sql" "$DB_NAME_backup_${TIMESTAMP}.zip"
fi
```

```
student@student-HP-Pro-T... x student@student-HP-Pro-T...
GNU nano 7.2 filename.s
#!/bin/bash

if [ "$EUID" -eq 0 ]
then
    echo "You are the root user."
else
    echo "You are NOT the root user."
fi
```

CONCLUSION:

This assignment provides a clear understanding of shell scripting and its importance in the Linux operating system. By writing and executing shell programs, students gain practical knowledge of automating system tasks and managing system resources effectively. Shell scripting is an essential skill for IT professionals, and this assignment helps build a strong foundation for future work in system administration and automation.